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1. Introduction

1.1 Background

UART (Universal Asynchronous Receiver/Transmitter) is a widely used serial communication interface for transferring data between digital systems. It provides asynchronous communication, meaning that a separate clock signal is not transmitted between the communicating devices. Instead, the transmitter and receiver operate using a predefined baud rate. In a UART system, parallel data is converted into a serial bit stream during transmission and converted back into parallel data at the receiver. A typical 8-bit UART frame consists of a start bit, eight data bits, and a stop bit, with the data transmitted LSB first.

1.2 Project Overview

This project focuses on the design and implementation of an **8-bit UART system using Verilog HDL**. The design is divided into four main RTL blocks: a **baud-rate generator**, **UART transmitter**, **UART receiver**, and **top-level UART module**.

The baud-rate generator produces the timing tick required for UART communication. The transmitter converts 8-bit parallel data into a serial UART frame, while the receiver reconstructs the original 8-bit data from the received serial stream. The top-level module integrates these blocks into a complete UART communication system.

The transmitter follows the FSM sequence:

IDLE → START → DATA → STOP

while the receiver follows:

IDLE → START → DATA → STOP → DONE

1.3 UART Operation and Verification

During transmission, the UART line remains HIGH when it is idle. A transmission begins with a LOW start bit, followed by eight data bits transmitted LSB first, and ends with a HIGH stop bit. The receiver detects the start bit, samples the incoming data bits, checks the stop bit, and indicates the completion of reception using the `rx_done` signal.

Functional verification is performed using individual testbenches for the baud-rate generator, transmitter, receiver, and top-level UART. Different data patterns, including `8'h55`, `8'hAA`, `8'h00`, `8'hFF`, `8'hA5`, and `8'h3C`, are used to verify correct transmission and reception. The transmitted and received data are compared to ensure that the UART operates correctly.

1.4 Simulation and Synthesis

The developed RTL design is first verified through simulation before proceeding to synthesis. Simulation waveforms are analyzed to observe the behavior of important signals such as `baud_tick`, `tx`, `rx`, `tx_data`, `rx_data`, `busy`, and `rx_done`. This helps verify the correct operation of the UART frame and the different FSM states. After successful functional verification, the complete UART design is synthesized using **Synopsys Design Compiler**. The `uart_top` module is selected as the top-level design, with the baud-rate generator, transmitter, and receiver forming its internal hierarchy. The synthesized design is then evaluated using area, timing, power, and cell-utilization reports.

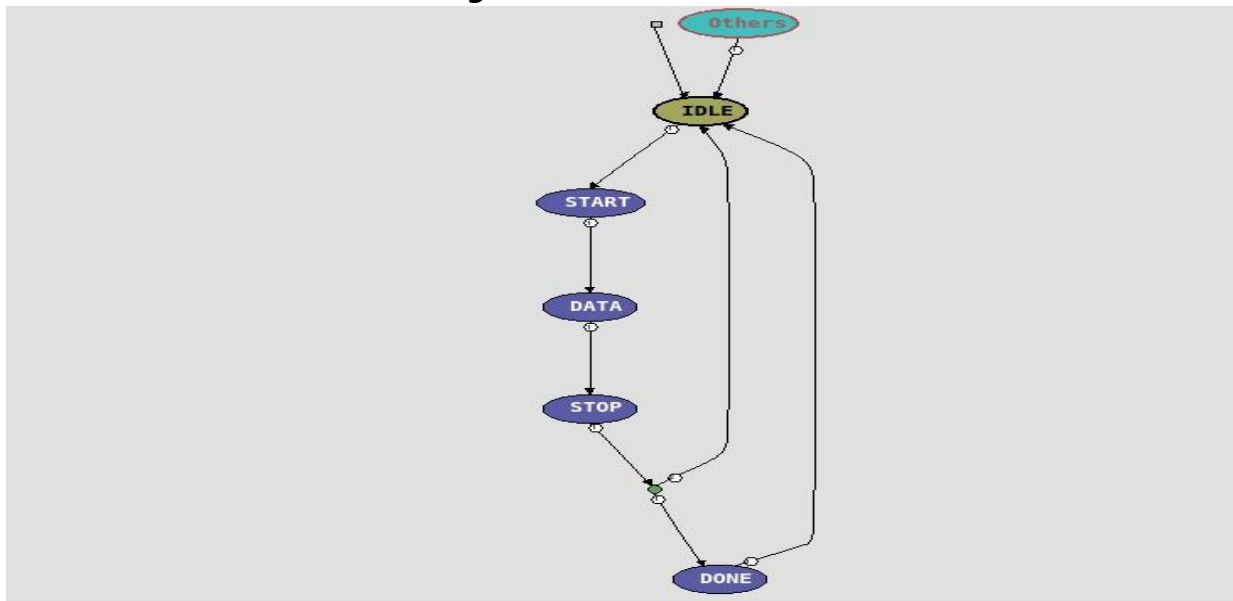
1.5 Project Scope and Objectives

The scope of this project covers the complete RTL-to-synthesis flow of an 8-bit UART system. The work begins with the development of individual RTL modules and their integration into a top-level UART design. Functional

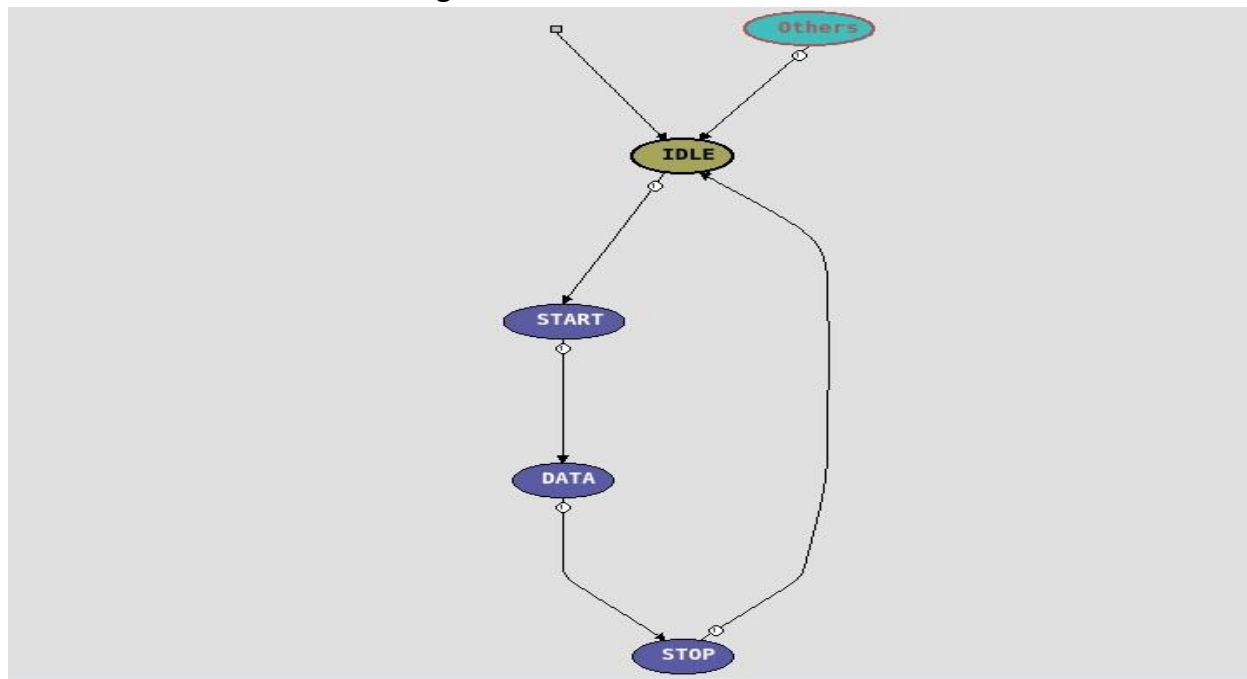
correctness is established through simulation and waveform analysis using dedicated testbenches. The verified design is then taken through synthesis using a standard-cell technology library, followed by analysis of its implementation characteristics. The primary objective is to develop a functionally correct and synthesizable UART system while gaining practical understanding of RTL design, finite-state-machine implementation, functional verification, waveform analysis, synthesis, timing constraints, and post-synthesis analysis. The project therefore demonstrates the complete digital design flow from **RTL development and verification to synthesis and timing evaluation**.

2. Block Diagram

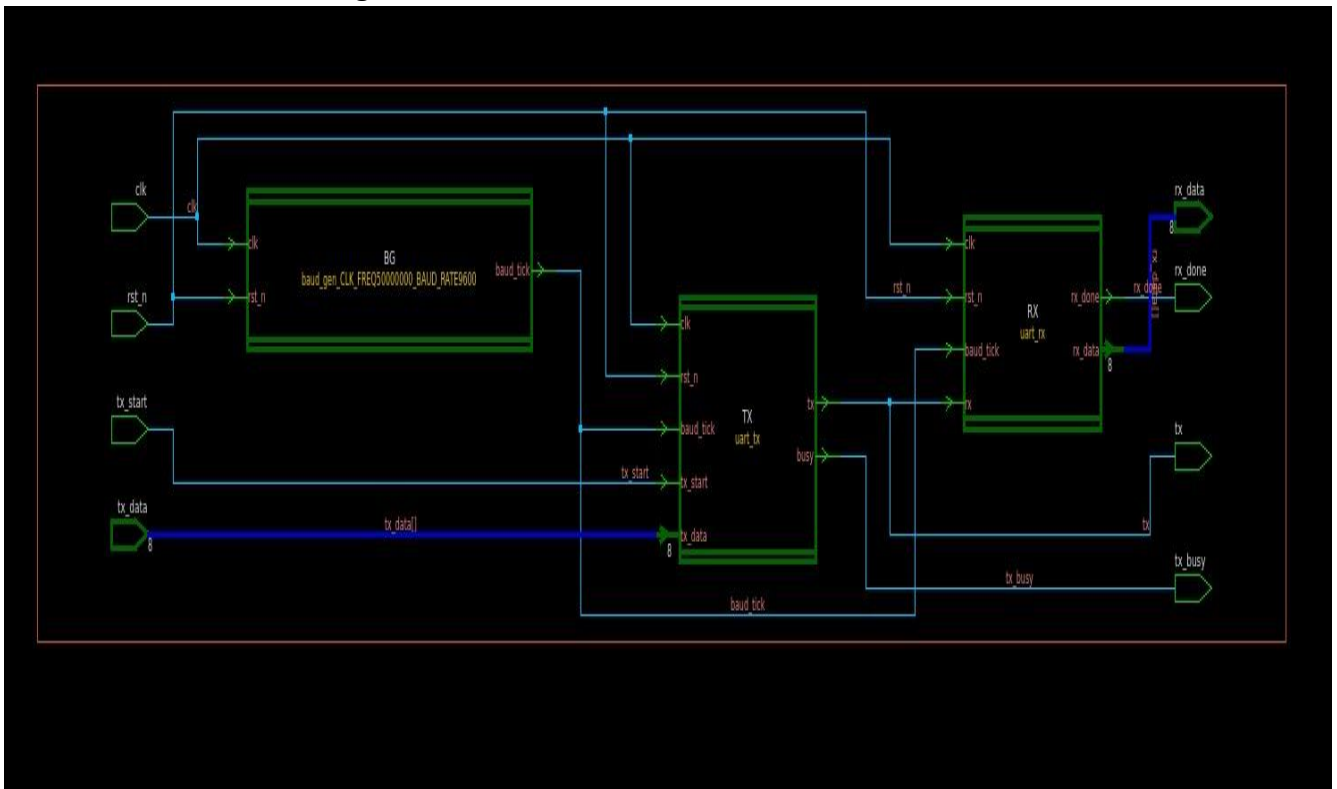
a. UART receiver FSM diagram



b. UART transmitter FSM diagram



c. UART Schematic diagram



3. Modules and codes:

I. Baud tick generator module

```

baud_gen #(
    parameter CLK_FREQ = 50000000, // 50 MHz
    parameter BAUD_RATE = 9600
)(
    input  clk,      input
    rst_n,          output reg
    baud_tick
);

```

```

    // Clock cycles required for one baud period
    localparam BAUD_DIV = CLK_FREQ / BAUD_RATE;

```

```

    // Counter width = 13 bits for BAUD_DIV = 434
    reg [14:0] count;

```

```

    always @(posedge clk or negedge rst_n)
    begin
        if (!rst_n)

```

```

        begin          count  <= 15'd0;
    baud_tick <= 1'b0;    end      else
    begin          if (count == BAUD_DIV-
    1)              begin          count
    <= 15'd0;          baud_tick <=
    1'b1;              end          else
    begin          count  <= count
    + 1'b1;          baud_tick <= 1'b0;
    end      end      end

```

```

endmodule

```

II. RECEIVER

```

module uart_rx(
input clk,    input rst_n,
input baud_tick,    input
rx,    output reg [7:0]
rx_data,    output reg
rx_done
);
    parameter IDLE = 3'd0;    parameter
START = 3'd1;    parameter DATA = 3'd2;
parameter STOP = 3'd3;    parameter
DONE = 3'd4;    reg [2:0] state;    reg
[2:0] bit_count;    reg [7:0] shift_reg;
always @(posedge clk or negedge rst_n)
begin
    if(!rst_n)
        begin          state
    <= IDLE;          bit_count
    <= 3'd0;          shift_reg
    <= 8'd0;          rx_data
    <= 8'd0;          rx_done
    <= 1'b0;    end      else
    begin          rx_done <=
    1'b0;

case(state)

//-----
// IDLE

```

```

//-----
-      IDLE:      begin
if(rx == 1'b0)      state <=
START;      end

//-----
// START BIT
//-----
-      START:      begin
if(baud_tick)      begin
bit_count <= 3'd0;      state <=
DATA;      end      end
//-----
// RECEIVE DATA      //-----
-----
DATA:      begin
if(baud_tick)      begin
shift_reg[bit_count] <= rx;

if(bit_count == 3'd7)
state <= STOP;

bit_count <= bit_count +
1'b1;      end      end
//-----
// STOP BIT
//-----
-      STOP:      begin
if(baud_tick)      begin
if(rx == 1'b1)      begin
rx_data <= shift_reg;
state <= DONE;      end
else      begin
state <= IDLE;      end
end      end
//-----
// DONE
//-----
-      DONE:      begin

```

```

rx_done <= 1'b1;          state <=
IDLE;          end
          //-----
// DEFAULT
          //-----
-          default:          begin
state <= IDLE;
          end
endcase
end    end
endmodule

```

III. TRANSMITTER

```

module uart_tx (
input clk,    input
rst_n,    input
baud_tick,    input
tx_start,    input
[7:0] tx_data,

output reg tx,
output reg busy
);

parameter IDLE = 2'b00;
parameter START = 2'b01;
parameter DATA = 2'b10;
parameter STOP = 2'b11;

reg [1:0] state;
reg [7:0] shift_reg;
reg [2:0] bit_count;

always @(posedge clk or negedge rst_n) begin

    if (!rst_n) begin
state    <= IDLE;
shift_reg <= 8'h00;
bit_count <= 3'd0;
tx    <= 1'b1;

```

```

busy    <= 1'b0;
end      else begin
case (state)
IDLE: begin          tx
    <= 1'b1;          busy
    <= 1'b0;

                if (tx_start) begin
shift_reg    <= tx_data;
bit_count <= 3'd0;          busy
    <= 1'b1;          state <=
START;          end          end

                START: begin
tx <= 1'b0;          if
(baud_tick) begin
state <= DATA;          tx
    <= shift_reg[0];          end
end

                DATA: begin          if (baud_tick)
begin          if (bit_count == 3'd7) begin
state <= STOP;          tx <= 1'b1;
end          else begin
bit_count    <= bit_count    +    1'b1;
shift_reg <= shift_reg >> 1;          tx
    <= shift_reg[1];          end

end

                end          STOP:
begin          tx <= 1'b1;
if (baud_tick) begin
state <= IDLE;          busy
    <= 1'b0;          end
end          default: begin
                state <= IDLE;

tx    <= 1'b1;
busy <= 1'b0;
end          endcase
end      end
endmodule

```


IV. TOP

```
module uart_top(
    input clk,    input
rst_n,    input tx_start,
input [7:0] tx_data,
output [7:0] rx_data,
output rx_done,
output tx,    output
tx_busy
);
wire baud_tick; wire
serial_line;
////////////////////////////////////
// Baud Generator
//////////////////////////////////// baud_gen
#(
    .CLK_FREQ(50000000),
    .BAUD_RATE(9600)
)
BG
(
    .clk(clk),
    .rst_n(rst_n),
    .baud_tick(baud_tick)
);
////////////////////////////////////
// UART Transmitter
//////////////////////////////////// uart_tx
TX
(
    .clk(clk),
    .rst_n(rst_n),
    .baud_tick(baud_tick),
    .tx_start(tx_start),
    .tx_data(tx_data),
    .tx(serial_line),
    .busy(tx_busy)
);
////////////////////////////////////
// UART Receiver
```

```

////////////////////////////////////// uart_rx
RX
(
    .clk(clk),
    .rst_n(rst_n),
    .baud_tick(baud_tick),
    .rx(serial_line),
    .rx_data(rx_data),
    .rx_done(rx_done)
);
assign tx = serial_line; endmodule

```

4. Testbench code

I. Baud tick generator

```

`timescale 1ns/1ps module
tb_baud_gen; // Parameters
parameter CLK_FREQ = 50_000_000;
parameter BAUD_RATE = 9600; //
Testbench Signals reg clk; reg
rst_n; wire baud_tick; // DUT
Instantiation baud_gen #(
    .CLK_FREQ(CLK_FREQ),
    .BAUD_RATE(BAUD_RATE)
) DUT (
    .clk(clk),
    .rst_n(rst_n),
    .baud_tick(baud_tick)
);
//=====
// Clock Generation (50 MHz)
//=====
initial begin    clk = 0;
    forever #10 clk = ~clk; // 20 ns period
end //=====
// Test Stimulus
//=====
initial begin    rst_n = 0    #100;    rst_n = 1
    // Run simulation long enough to observe
    // several baud_tick pulses

```

```

        #100000;
$finish; end
initial begin
    $display("Time\tReset\tBaud Tick");
    $monitor("%0t\t%b\t%b", $time, rst_n, baud_tick);
end initial begin
    $dumpfile("baud_gen.fsdb");
    $dumpvars(0, tb_baud_gen);
end endmodule

```

II. RECEIVER TEST BENCH

```

module tb_uart_rx;

//=====

// Testbench Signals

//=====

reg clk; reg rst_n; reg baud_tick; reg rx;

wire [7:0] rx_data;
wire rx_done;
//=====
=====
=====
=====

// Instantiate DUT
//=====
uart_rx DUT (
    .clk(clk),
    .rst_n(rst_n),
    .baud_tick(baud_tick),
    .rx(rx),
    .rx_data(rx_data),
    .rx_done(rx_done)
);
//=====

// Clock Generation (50 MHz)
//=====

initial begin    clk = 0;
    forever #10 clk = ~clk;
end

//=====

```

```

// Baud Tick Generation

//=====
initial begin      baud_tick = 0;      forever begin
    #80 baud_tick =
1;      #20 baud_tick
= 0;      end      end
//=====
// UART Byte Transmission Task

//=====
task send_byte;      input [7:0] data;      integer i;      begin
    // Idle
rx = 1;
    @(posedge baud_tick);
    // Start Bit
rx = 0;
    @(posedge
baud_tick);
    // Data Bits (LSB
First)
    for(i=0; i<8; i=i+1)
begin          rx
= data[i];
        @(posedge
baud_tick);      end          //
Stop Bit          rx = 1;
        @(posedge
baud_tick);      end
endtask
//=====
// Stimulus

//=====
initial begin      rst_n = 0;      rx  = 1;      #100;      rst_n = 1;
send_byte(8'hA5);
    #500;
send_byte(8'h3C);
    #500;
send_byte(8'hF0);

```

```

        #1000;
$finish;    end

//=====
// Monitor
//=====

initial begin
    $monitor("Time=%0t  RX=%b  RX_DATA=%h  RX_DONE=%b",
        $time, rx, rx_data, rx_done);
end

//=====
// Waveform Dump
//=====

initial begin
    $dumpfile("uart_rx.vcd");
    $dumpvars(0,
tb_uart_rx);    end    initial
begin
    $dumpfile("uart_rx.fsdb");
    $dumpvars(1);
end
endmodule

```

III. TRANSMITTER TEST BENCH

```

`timescale 1ns/1ps

module tb_uart_tx;

//=====
// Parameters
//=====

parameter CLK_PERIOD = 20;

//=====
// Testbench Signals

//=====

reg clk;    reg rst_n;    reg baud_tick;    reg tx_start;    reg [7:0] tx_data;    wire tx;
wire busy;

//=====
// Instantiate DUT
//=====

uart_tx DUT (
    .clk(clk),

```

```

        .rst_n(rst_n),
        .baud_tick(baud_tick),
        .tx_start(tx_start),
        .tx_data(tx_data),
        .tx(tx),
        .busy(busy)
    );
//=====
    // Clock Generation (50 MHz)
//=====
always #(CLK_PERIOD/2) clk = ~clk;
//=====

    // Baud Tick Generation

//=====
initial begin    baud_tick = 0;    forever begin
    #80 baud_tick = 1;
#20 baud_tick = 0;    end
end //=====
    // Stimulus

//=====
initial begin    clk    = 0;    rst_n    = 0;    tx_start = 0;    tx_data = 8'h00;
    #100;
rst_n = 1;
    //-----
    // Test Case 1
    //-----
    #50;    tx_data
= 8'hA5;    tx_start
= 1;        #20
tx_start    =    0;
wait(busy == 0);
    //-----
    // Test Case 2
    //-----
    #200    tx_data
= 8'h3C;    tx_start
= 1;        #20;

```

```

tx_start    =    0;
wait(busy == 0);
    //-----
    // Test Case 3
    //-----
    #200;
tx_data     =  8'hFF;
tx_start = 1;    #20;
tx_start    =    0;
wait(busy == 0);
    #500;
$finish;    end
//=====
    // Monitor
//=====
initial begin
    $monitor("Time=%0t | TX=%b | Busy=%b | Data=%h",
        $time, tx, busy, tx_data);
end
//=====
    // Dump Waveform
//=====
initial begin
    $dumpfile("uart_tx.vcd");
    $dumpvars(0, tb_uart_tx);
end    initial begin
    $dumpfile("uart_tx.fsdb");
    $dumpvars(1);
end
endmodule

```

IV. TOP MODULE TEST BENCH

```

`timescale 1ns/1ps module
tb_uart_top;
    reg    clk;    reg
rst_n;    reg
tx_start;    reg [7:0]
tx_data;    wire
tx;    wire
tx_busy;    wire [7:0]
rx_data;    wire
rx_done;    integer
pass_count;    integer

```

```

fail_count;
//=====
=====
=====
=====

// DUT
//=====
uart_top DUT (
    .clk    (clk),
    .rst_n  (rst_n),
    .tx_start (tx_start),
    .tx_data (tx_data),
    .tx      (tx),
    .tx_busy (tx_busy),
    .rx_data (rx_data),
    .rx_done (rx_done)
);
//=====
// Clock Generation
// 50 MHz -> 20 ns period
//=====
initial begin    clk = 1'b0;    forever #10 clk = ~clk;    end
//=====
// Test Task
//=====
task send_and_check;    input [7:0] data;    begin
    // Wait until transmitter is free
    wait(tx_busy == 1'b0);

    // Apply data
    @(posedge    clk);
    tx_data      =    data;
    tx_start     =    1'b1;
    @(posedge    clk);
    tx_start = 1'b0;

    // Wait for receiver to finish
    wait(rx_done == 1'b1);    //
    Check received data
    if(rx_data == data)    begin
        $display(
            "PASS: TX_DATA = %h, RX_DATA = %h",
            data,            rx_data
        );
    end
end

```



```

        pass_count = pass_count + 1;
end    else    begin
$display(
    "FAIL: TX_DATA = %h, RX_DATA = %h"
data,    rx_data
    );
    fail_count = fail_count + 1;
end
    // Allow rx_done to return low
    @(posedge clk);
end    endtask

//=====
// Test Sequence
//=====
initial begin    // Initialize    rst_n    = 1'b0;    tx_start = 1'b0;
tx_data = 8'h00;    pass_count = 0;    fail_count = 0;
    // Reset
#100;    rst_n
= 1'b1;

    #100
send_and_check(8'h55);
send_and_check(8'hAA);
send_and_check(8'h00);
send_and_check(8'hFF);
send_and_check(8'hA5);
send_and_check(8'h3C);

    //=====
    // Final Result
    //=====
#500;
    $display("-----");
    $display("UART TEST COMPLETE");
    $display("PASSED = %0d", pass_count);
    $display("FAILED = %0d", fail_count);
    $display("-----");
if(fail_count == 0)
    $display("ALL TESTS PASSED");
else
    $display("SOME TESTS FAILED")
    $finish;
end    initial
begin
    $dumpfile("uart_top.vcd");
$dumpvars(0, tb_uart_top);
end    initial begin
    $dumpfile("uart_top.fsdb");

```

```

    $dumpvars(1);
end
endmodule

```

5. Wave forms

a. UART Baud tick generator

```

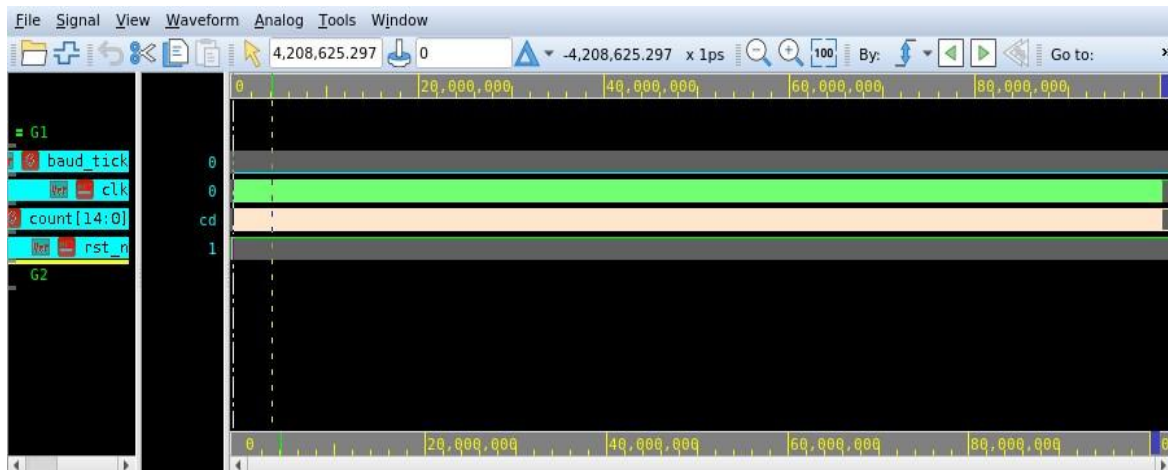
output.log (/home/iter_work/iter...na_dash/UART_synthesis) - GVIM11
File Edit Tools Syntax Buffers Window Help

Command: /home/iter_work/iter4/upaasana_dash/UART_synthesis/./simv +vcs+lic+wait -l output.log
Chronologic VCS simulator copyright 1991-2024
Contains Synopsys proprietary information.
Compiler version W-2024.09-SP2_Full64; Runtime version W-2024.09-SP2_Full64; Aug 8 10:17 2026
Time      Reset    Baud Tick
0          0         0
100000    1          0
$finish called from file "tb_baud_gen.v", line 46.
$finish at simulation time      100100000
V C S   S i m u l a t i o n   R e p o r t
Time: 100100000 ps
CPU Time:      0.580 seconds;      Data structure size:      0.0Mb
Sat Aug 8 10:17:35 2026

~
~
~
~
~
~

"output.log" 13L, 581C                                1,1                                All

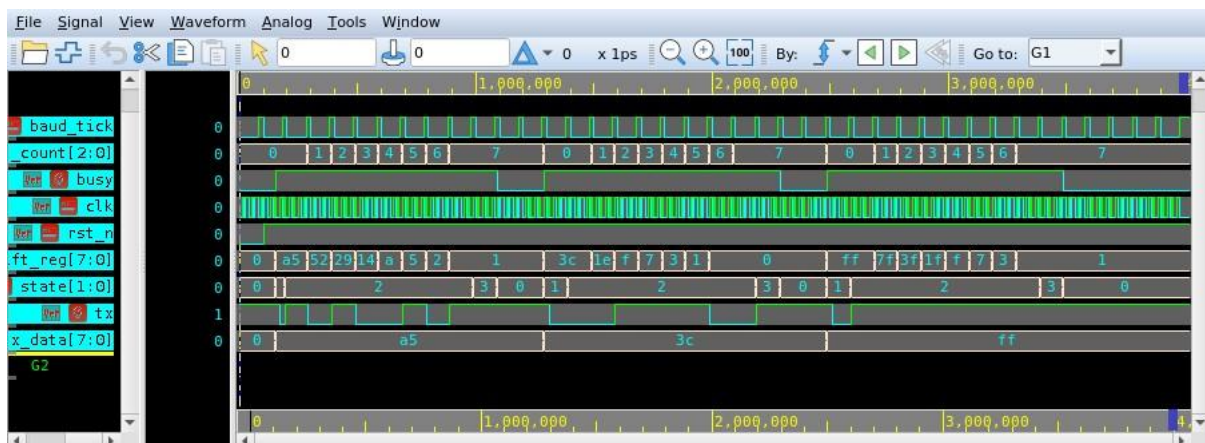
```



b. UART transmitter

```
output.log (/home/iter_work/iter...ana_dash/UART_synthesis) - GVIM9
File Edit Tools Syntax Buffers Window Help

Command: /home/iter_work/iter4/upaasana_dash/UART_synthesis/./simv +vcs+lic+wait -l output.log
Chronologic VCS simulator copyright 1991-2024
Contains Synopsys proprietary information.
Compiler version W-2024.09-SP2_Full64; Runtime version W-2024.09-SP2_Full64; Aug 8 10:02 2026
Time=0 | TX=1 | Busy=0 | Data=00
Time=150000 | TX=1 | Busy=1 | Data=a5
Time=170000 | TX=0 | Busy=1 | Data=a5
Time=190000 | TX=1 | Busy=1 | Data=a5
Time=290000 | TX=0 | Busy=1 | Data=a5
Time=390000 | TX=1 | Busy=1 | Data=a5
Time=490000 | TX=0 | Busy=1 | Data=a5
Time=690000 | TX=1 | Busy=1 | Data=a5
Time=790000 | TX=0 | Busy=1 | Data=a5
Time=890000 | TX=1 | Busy=1 | Data=a5
Time=1090000 | TX=1 | Busy=0 | Data=a5
Time=1290000 | TX=1 | Busy=1 | Data=3c
Time=1310000 | TX=0 | Busy=1 | Data=3c
Time=1590000 | TX=1 | Busy=1 | Data=3c
Time=1990000 | TX=0 | Busy=1 | Data=3c
Time=2190000 | TX=1 | Busy=1 | Data=3c
Time=2290000 | TX=1 | Busy=0 | Data=3c
Time=2490000 | TX=1 | Busy=1 | Data=ff
Time=2510000 | TX=0 | Busy=1 | Data=ff
Time=2590000 | TX=1 | Busy=1 | Data=ff
Time=3490000 | TX=1 | Busy=0 | Data=ff
$finish called from file "tb_uart_tx.v", line 104.
$finish at simulation time 3990000
VCS Simulation Report
Time: 3990000 ps
CPU Time: 0.610 seconds; Data structure size: 0.0Mb
Sat Aug 8 10:02:29 2026
~
~
~
```



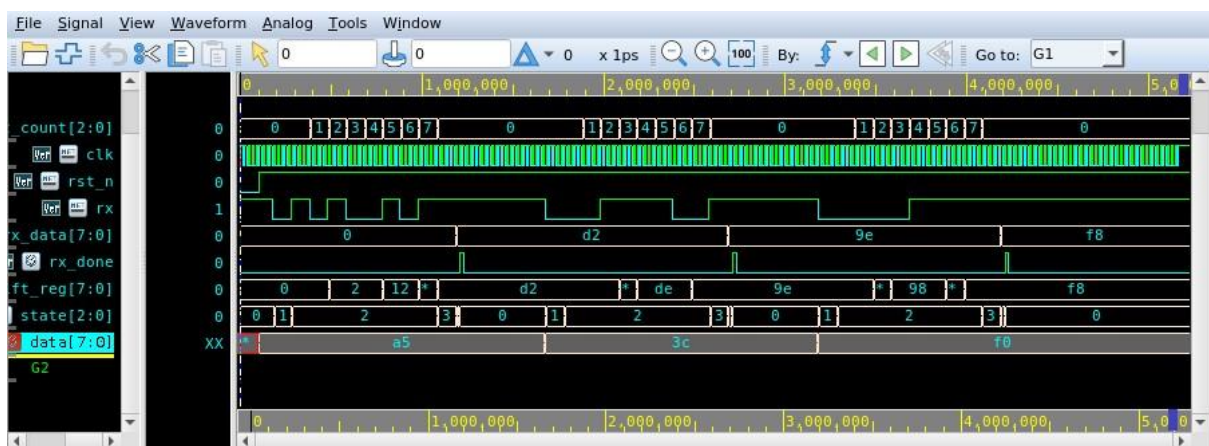
c. UART receiver

```
output.log (/home/iter_work/iter...na_dash/UART_synthesis) - GVIM11
File Edit Tools Syntax Buffers Window Help

Command: /home/iter_work/iter4/upaasana_dash/UART_synthesis/./simv +vcs+lic+wait -l output.log
Chronologic VCS simulator copyright 1991-2024
Contains Synopsys proprietary information.
Compiler version W-2024.09-SP2_Full64; Runtime version W-2024.09-SP2_Full64; Aug 8 10:14 2026
Time=0 RX=1 RX_DATA=00 RX_DONE=0
Time=180000 RX=0 RX_DATA=00 RX_DONE=0
Time=280000 RX=1 RX_DATA=00 RX_DONE=0
Time=380000 RX=0 RX_DATA=00 RX_DONE=0
Time=480000 RX=1 RX_DATA=00 RX_DONE=0
Time=580000 RX=0 RX_DATA=00 RX_DONE=0
Time=780000 RX=1 RX_DATA=00 RX_DONE=0
Time=880000 RX=0 RX_DATA=00 RX_DONE=0
Time=980000 RX=1 RX_DATA=00 RX_DONE=0
Time=1190000 RX=1 RX_DATA=d2 RX_DONE=0
Time=1210000 RX=1 RX_DATA=d2 RX_DONE=1
Time=1230000 RX=1 RX_DATA=d2 RX_DONE=0
Time=1680000 RX=0 RX_DATA=d2 RX_DONE=0
Time=1980000 RX=1 RX_DATA=d2 RX_DONE=0
Time=2380000 RX=0 RX_DATA=d2 RX_DONE=0
Time=2580000 RX=1 RX_DATA=d2 RX_DONE=0
Time=2690000 RX=1 RX_DATA=9e RX_DONE=0
Time=2710000 RX=1 RX_DATA=9e RX_DONE=1
Time=2730000 RX=1 RX_DATA=9e RX_DONE=0
Time=3180000 RX=0 RX_DATA=9e RX_DONE=0
Time=3680000 RX=1 RX_DATA=9e RX_DONE=0
Time=4190000 RX=1 RX_DATA=f8 RX_DONE=0
Time=4210000 RX=1 RX_DATA=f8 RX_DONE=1
Time=4230000 RX=1 RX_DATA=f8 RX_DONE=0
$finish called from file "tb_uart_rx.v", line 101.
$finish at simulation time 5180000
VCS Simulation Report
Time: 5180000 ps
CPU Time: 0.590 seconds; Data structure size: 0.0Mb
Sat Aug 8 10:15:00 2026
~
~
~
```

1,1

All



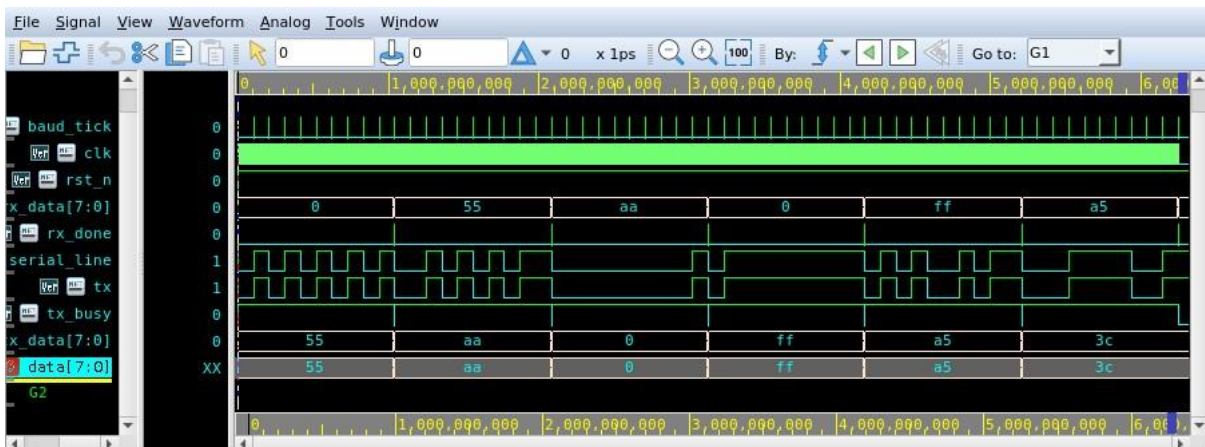
d. Whole UART module (top module)

```

output.log (/home/iter_work/iter...na_dash/UART_sysnthesis) - GVIM10
File Edit Tools Syntax Buffers Window Help

Command: /home/iter_work/iter4/upaasana_dash/UART_sysnthesis/./simv +vcs+lic+wait -l output.log
Chronologic VCS simulator copyright 1991-2024
Contains Synopsys proprietary information.
Compiler version W-2024.09-SP2_Full64; Runtime version W-2024.09-SP2_Full64; Aug 8 10:10 2026
PASS: TX_DATA = 55, RX_DATA = 55
PASS: TX_DATA = aa, RX_DATA = aa
PASS: TX_DATA = 00, RX_DATA = 00
PASS: TX_DATA = ff, RX_DATA = ff
PASS: TX_DATA = a5, RX_DATA = a5
PASS: TX_DATA = 3c, RX_DATA = 3c
-----
UART TEST COMPLETE
PASSED = 6
FAILED = 0
-----
ALL TESTS PASSED
$finish called from file "tb_uart_top.v", line 144.
$finish at simulation time 6250250000
VCS Simulation Report
Time: 6250250000 ps
CPU Time: 0.620 seconds; Data structure size: 0.0Mb
Sat Aug 8 10:10:08 2026
~

```



6. Synthesis scripts and reports

```

dc_shell> set link_library {*}
/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db} *
/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db
dc_shell> set target_library
{/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db}
/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db dc_shell>
analyze -library WORK -format verilog
/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/baud_gen.v
Running PRESTO HDLC
Compiling source file

```


/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/ baud_gen.v Presto
compilation completed successfully.

1

dc_shell> analyze -library WORK -format verilog
/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_tx.v
Running PRESTO HDLC
Compiling source file
/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_tx.v Presto
compilation completed successfully.

1

dc_shell> analyze -library WORK -format verilog
/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_rx.v
Running PRESTO HDLC
Compiling source file
/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_rx.v Presto
compilation completed successfully.

1

dc_shell> analyze -library WORK -format verilog
/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_top.v
Running PRESTO HDLC
Compiling source file
/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_top.v Presto
compilation completed successfully.

1

dc_shell> elaborate uart_top
Running PRESTO HDLC
Presto compilation completed successfully. (uart_top)
Module: uart_top, Ports: 22, Input: 11, Output: 11, Inout: 0
Module: uart_top, Registers: 0, Async set/reset: 0, Sync set/reset: 0
Information: Module uart_top report end. (ELAB-965)
Warning: Overwriting design file
'/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_top.db'. (DDB-24)
Elaborated 1 design.
Current design is now 'uart_top'.
Reading ddc file
'/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/ baud_gen_CLK_FREQ5000
0000_BAUD_RATE9600.ddc'.
Reading ddc file
'/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_tx.ddc'.
Reading ddc file
'/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_rx.ddc'.

1

Current design is 'uart_top'.

dc_shell> get_ports

```
{tx_data[7] tx_data[6] tx_data[5] tx_data[4] tx_data[3] tx_data[2] tx_data[1] tx_data[0] rx_data[7]
rx_data[6] rx_data[5] rx_data[4] rx_data[3] rx_data[2] rx_data[1] rx_data[0] clk rst_n tx_start
rx_done tx tx_busy} dc_shell> get_pins
```

```
{BG/clock BG/rst_n BG/baud_tick TX/clock TX/rst_n TX/baud_tick TX/tx_start TX/tx_data[7]
TX/tx_data[6] TX/tx_data[5] TX/tx_data[4] TX/tx_data[3] TX/tx_data[2] TX/tx_data[1]
TX/tx_data[0] TX/tx_busy RX/clock RX/rst_n RX/baud_tick RX/rx RX/rx_data[7] RX/rx_data[6]
RX/rx_data[5] RX/rx_data[4] RX/rx_data[3] RX/rx_data[2] RX/rx_data[1] RX/rx_data[0]
RX/rx_done} dc_shell> current_design
```

Current design is 'uart_top'.

{uart_top}

dc_shell> create_clock -name clk -period 2

Warning: Creating virtual clock named 'clk' with no sources. (UID-348)

1

dc_shell> get_lib

{saed32rvt_tt0p85v25c gtech standard.sldb dw_foundation.sldb} dc_shell>

get_clocks

{clk} dc_shell>

start_gui

dc_shell> create_clock -name clk -period 2

Warning: Creating virtual clock named 'clk' with no sources. (UID-348)

1

dc_shell> get_clocks

{clk}

dc_shell> set_input_delay 1 -clock clk [all_inputs]

1

dc_shell> set_output_delay 1 -clock clk [all_outputs]

1

dc_shell> link

Linking design 'uart_top'

Using the following designs and libraries:

* (4 designs)

/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/uart_top.db, etc

saed32rvt_tt0p85v25c (library)

/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db

1

1

dc_shell> check_timing

Information: Changed wire load model for 'uart_rx' from '(none)' to 'ForQA'. (OPT-170)

Information: Changed wire load model for 'uart_tx' from '(none)' to 'ForQA'. (OPT-170)

Information: Changed wire load model for 'baud_gen_CLK_FREQ50000000_BAUD_RATE9600' from '(none)' to 'ForQA'. (OPT-170)

Module: DW01_inc_width15, Ports: 30, Input: 15, Output: 15, Inout: 0

Module: DW01_inc_width15, Registers: 0, Async set/reset: 0, Sync set/reset: 0

Module: DW01_inc_width3, Ports: 6, Input: 3, Output: 3, Inout: 0

Module: DW01_inc_width3, Registers: 0, Async set/reset: 0, Sync set/reset: 0

Information: Changed wire load model for 'DW01_inc_width3' from '(none)' to 'ForQA'. (OPT170)

Information: Changed wire load model for '*ADD_UNNS_OP_3_1_3' from '(none)' to 'ForQA'. (OPT-170)

Information: Changed wire load model for 'DW01_inc_width3' from '(none)' to 'ForQA'. (OPT170)

Information: Changed wire load model for '*ADD_UNNS_OP_3_1_3' from '(none)' to 'ForQA'. (OPT-170)

Information: Changed wire load model for 'DW01_inc_width15' from '(none)' to 'ForQA'. (OPT170)

Information: Changed wire load model for '*ADD_UNNS_OP_15_1_15' from '(none)' to 'ForQA'. (OPT-170)

Information: Updating design information... (UID-85)

Information: Checking generated_clocks...

Information: Checking loops...

Information: Checking no_input_delay...

Information: Checking unconstrained_endpoints...

Warning: The following end-points are not constrained for maximum delay.

End point

BG/ baud_tick_reg/next_state

BG/ count_reg[0]/next_state

BG/ count_reg[1]/next_state

BG/ count_reg[2]/next_state

BG/ count_reg[3]/next_state

BG/count_reg[4]/next_state
BG/count_reg[5]/next_state
BG/count_reg[6]/next_state
BG/count_reg[7]/next_state
BG/count_reg[8]/next_state
BG/count_reg[9]/next_state
BG/count_reg[10]/next_state
BG/count_reg[11]/next_state
BG/count_reg[12]/next_state
BG/count_reg[13]/next_state
BG/count_reg[14]/next_state
RX/bit_count_reg[0]/next_state
RX/bit_count_reg[0]/synch_enable
RX/bit_count_reg[1]/next_state
RX/bit_count_reg[1]/synch_enable
RX/bit_count_reg[2]/next_state
RX/bit_count_reg[2]/synch_enable
RX/rx_data_reg[0]/next_state
RX/rx_data_reg[0]/synch_enable
RX/rx_data_reg[1]/next_state
RX/rx_data_reg[1]/synch_enable
RX/rx_data_reg[2]/next_state
RX/rx_data_reg[2]/synch_enable
RX/rx_data_reg[3]/next_state
RX/rx_data_reg[3]/synch_enable
RX/rx_data_reg[4]/next_state
RX/rx_data_reg[4]/synch_enable
RX/rx_data_reg[5]/next_state
RX/rx_data_reg[5]/synch_enable
RX/rx_data_reg[6]/next_state
RX/rx_data_reg[6]/synch_enable
RX/rx_data_reg[7]/next_state
RX/rx_data_reg[7]/synch_enable
RX/rx_done_reg/next_state
RX/shift_reg_reg[0]/next_state
RX/shift_reg_reg[0]/synch_enable
RX/shift_reg_reg[1]/next_state
RX/shift_reg_reg[1]/synch_enable
RX/shift_reg_reg[2]/next_state
RX/shift_reg_reg[2]/synch_enable

RX/shift_reg_reg[3]/next_state
RX/shift_reg_reg[3]/synch_enable
RX/shift_reg_reg[4]/next_state
RX/shift_reg_reg[4]/synch_enable
RX/shift_reg_reg[5]/next_state
RX/shift_reg_reg[5]/synch_enable
RX/shift_reg_reg[6]/next_state
RX/shift_reg_reg[6]/synch_enable
RX/shift_reg_reg[7]/next_state
RX/shift_reg_reg[7]/synch_enable RX/state_reg[0]/next_state
RX/state_reg[0]/synch_enable
RX/state_reg[1]/next_state
RX/state_reg[1]/synch_enable
RX/state_reg[2]/next_state
RX/state_reg[2]/synch_enable
TX/bit_count_reg[0]/next_state
TX/bit_count_reg[0]/synch_enable
TX/bit_count_reg[1]/next_state
TX/bit_count_reg[1]/synch_enable
TX/bit_count_reg[2]/next_state
TX/bit_count_reg[2]/synch_enable
TX/busy_reg/next_state
TX/busy_reg/synch_enable
TX/shift_reg_reg[0]/next_state
TX/shift_reg_reg[0]/synch_enable
TX/shift_reg_reg[1]/next_state
TX/shift_reg_reg[1]/synch_enable
TX/shift_reg_reg[2]/next_state
TX/shift_reg_reg[2]/synch_enable
TX/shift_reg_reg[3]/next_state
TX/shift_reg_reg[3]/synch_enable
TX/shift_reg_reg[4]/next_state
TX/shift_reg_reg[4]/synch_enable
TX/shift_reg_reg[5]/next_state
TX/shift_reg_reg[5]/synch_enable
TX/shift_reg_reg[6]/next_state
TX/shift_reg_reg[6]/synch_enable
TX/shift_reg_reg[7]/next_state
TX/shift_reg_reg[7]/synch_enable
TX/state_reg[0]/next_state

TX/state_reg[0]/synch_enable
TX/state_reg[1]/next_state
TX/state_reg[1]/synch_enable
TX/tx_reg/next_state
TX/tx_reg/synch_enable
rx_data[0] rx_data[1]
rx_data[2] rx_data[3]

rx_data[4]
rx_data[5]
rx_data[6]
rx_data[7] rx_done
tx tx_busy

Information: Checking pulse_clock_cell_type...

Information: Checking no_driving_cell...

Information: Checking partial_input_delay...

1

Current design is 'uart_top'.

dc_shell> compile

CPU Load: 6%, Ram Free: 1 GB, Swap Free: 30 GB, Work Disk Free: 890 GB, Tmp Disk Free: 30 GB

Information: Evaluating DesignWare library utilization. (UISN-27)

=====			
=====			
DesignWare Building Block Library	Version	Available	
=====			
=====			
Basic DW Building Blocks	W-2024.09-DWBB_202409.3	*	
Licensed DW Building Blocks			
=====			
=====			

=====			
=====			
Flow Information			

Flow	Design Compiler		
=====			

=====	
Design Information	Value
=====	
Number of Scenarios	0
Leaf Cell Count	197
Number of User Hierarchies	3
Sequential Cell Count	54
Macro Count	0
Pad Count	0
Number of Power Domains	0
Number of Path Groups	2
Number of VT Class	1
Number of Clocks	1
Number of Dont Touch Cells	39
Number of Dont Touch Nets	0
Number of Size Only Cells	0
Design with Target Library Subset (with clock_path)	false (false)
Design with UPF Data	false
=====	
=====	

Information: There are 3 potential problems in your design. Please run 'check_design' for more information. (LINT-99)

Beginning Pass 1 Mapping

Processing 'uart_rx'

Module: DW01_inc_width3, Ports: 6, Input: 3, Output: 3, Inout: 0

Module: DW01_inc_width3, Registers: 0, Async set/reset: 0, Sync set/reset: 0

Processing 'uart_tx'

Processing 'baud_gen_CLK_FREQ50000000_BAUD_RATE9600'

Processing 'uart_top'

Updating timing information

Information: Updating design information... (UID-85)

Beginning Implementation Selection

Module: DW01_inc_width15, Ports: 30, Input: 15, Output: 15, Inout: 0

Module: DW01_inc_width15, Registers: 0, Async set/reset: 0, Sync set/reset: 0

Processing 'baud_gen_CLK_FREQ50000000_BAUD_RATE9600_DW01_inc_0_DW01_inc_2'

Beginning Mapping Optimizations (Medium effort)

		TOTAL			ENDPOINT
ELAPSED		WORST NEG	SETUP	DESIGN	
TIME	AREA	SLACK	COST	RULE COST	

0:00:02	879.8	0.00	0.0	1.9	
0:00:02	879.8	0.00	0.0	1.9	
0:00:02	879.8	0.00	0.0	1.9	
0:00:02	879.8	0.00	0.0	1.9	
0:00:02	879.8	0.00	0.0	1.9	
0:00:02	731.7	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	

Beginning Delay Optimization Phase

		TOTAL			ENDPOINT
ELAPSED		WORST NEG	SETUP	DESIGN	
TIME	AREA	SLACK	COST	RULE COST	

0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	
0:00:02	730.2	0.00	0.0	0.0	

Beginning Area-Recovery Phase (cleanup)

TOTAL					ENDPOINT	-----
ELAPSED	WORST	NEG	SETUP	DESIGN		
TIME	AREA	SLACK	COST	RULE COST		

0:00:02	730.2	0.00	0.0	0.0		
0:00:02	730.2	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		
0:00:03	725.1	0.00	0.0	0.0		

Loading db file
'/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db'

Note: Symbol # after min delay cost means estimated hold TNS across all active scenarios

Optimization Complete

CPU Load: 7%, Ram Free: 1 GB, Swap Free: 30 GB, Work Disk Free: 890 GB, Tmp Disk Free: 30 GB

Information: Total number of MV cells in the design.

MV Cells	Total Number

Level Shifter:	0
Enable Level Shifter:	0
Isolation Cell:	0
Retention Cell:	0
Retention Clamp Cell:	0
Switch Cell:	0

Always-On Cell: 0
Repeater Cell: 0

Unmapped MV Cells
----- 0
Isolation Cells are unmapped
0 Retention Clamp Cells are unmapped
----- 1
Current design is 'uart_top'.

dc_shell> compile_ultra
Loading db file '/home/Tools/synopsys/tool/install/syn/W-2024.09-SP3/libraries/syn/dw_foundation.sldb'
Warning: DesignWare synthetic library dw_foundation.sldb is added to the synthetic_library in the current command. (UISN-40)
Information: Performing leakage power optimization. (PWR-850)
CPU Load: 5%, Ram Free: 1 GB, Swap Free: 30 GB, Work Disk Free: 890 GB, Tmp Disk Free: 30 GB
Alib files are up-to-date.
Information: Evaluating DesignWare library utilization. (UISN-27)

=====

DesignWare Building Block Library		Version	Available
Basic DW Building Blocks		W-2024.09-DWBB_202409.3	*
Licensed DW Building Blocks		W-2024.09-DWBB_202409.3	*

=====

=====

Flow Information		
Flow	Design Compiler WLM	
Command Line	compile_ultra	

=====

Design Information		Value
--------------------	--	-------

=====

```

=====
| Number of Scenarios                | 0 |
| Leaf Cell Count                    | 210 |
| Number of User Hierarchies         | 4 |
| Sequential Cell Count              | 54 |
| Macro Count                        | 0 |
| Pad Count                          | 0 |
| Number of Power Domains            | 0 |
| Number of Path Groups              | 2 |
| Number of VT Class                 | 1 |
| Number of Clocks                   | 1 |
| Number of Dont Touch Cells         | 0 |
| Number of Dont Touch Nets          | 0 |
| Number of Size Only Cells          | 0 |
| Design with Target Library Subset (with clock_path) | false (false) |
| Design with UPF Data               | false |
=====
=====

```

Information: Sequential output inversion is enabled. SVF file must be used for formal verification. (OPT-1208)

Simplifying Design 'uart_top'

Loading db file

'/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db'

Loaded alib file './alib-52/saed32rvt_tt0p85v25c.db.alib'

Module: DW01_NAND2, Ports: 3, Input: 2, Output: 1, Inout: 0

Module: DW01_NAND2, Registers: 0, Async set/reset: 0, Sync set/reset: 0

Building model 'DW01_NAND2'

Information: Ungrouping hierarchy BG before Pass 1 (OPT-776)

Information: Ungrouping hierarchy TX before Pass 1 (OPT-776)

Information: Ungrouping hierarchy RX before Pass 1 (OPT-776)

Information: Ungrouping 3 of 5 hierarchies before Pass 1 (OPT-775)

CPU Load: 6%, Ram Free: 1 GB, Swap Free: 30 GB, Work Disk Free: 890 GB, Tmp Disk Free: 30 GB

Information: State dependent leakage is now switched from on to off.

Beginning Pass 1 Mapping

Processing 'uart_top'

Implement Synthetic for 'uart_top'.

Processing 'baud_gen_CLK_FREQ50000000_BAUD_RATE9600_DW01_inc_0_DW01_inc_2'

CPU Load: 6%, Ram Free: 1 GB, Swap Free: 30 GB, Work Disk Free: 890 GB, Tmp Disk Free: 30 GB

Updating timing information

Information: Updating design information... (UID-85)

Information: The library cell 'PMT3_RVT' in the library 'saed32rvt_tt0p85v25c' is not characterized for internal power. (PWR-536)

Information: The library cell 'PMT2_RVT' in the library 'saed32rvt_tt0p85v25c' is not characterized for internal power. (PWR-536)

Information: The library cell 'PMT1_RVT' in the library 'saed32rvt_tt0p85v25c' is not characterized for internal power. (PWR-536)

Information: The library cell 'NMT3_RVT' in the library 'saed32rvt_tt0p85v25c' is not characterized for internal power. (PWR-536)

Information: The library cell 'NMT2_RVT' in the library 'saed32rvt_tt0p85v25c' is not characterized for internal power. (PWR-536)

Information: The library cell 'NMT1_RVT' in the library 'saed32rvt_tt0p85v25c' is not characterized for internal power. (PWR-536)

Information: The target library(s) contains cell(s), other than black boxes, that are not characterized for internal power. (PWR-24)

Threshold voltage group cell usage:

>> saed32cell_svt 100.00%

Beginning Mapping Optimizations (Ultra High effort)

Information: There is no timing violation in design uart_top. Delay-based auto_ungroup will not be performed. (OPT-780)

TOTAL						
ELAPSED		WORST NEG	SETUP	DESIGN		LEAKAGE
TIME	AREA	SLACK	COST	RULE COST	ENDPOINT	POWER
0:00:04	808.9	0.00	0.0	0.0		38182644.0000
0:00:04	807.7	0.00	0.0	0.0		38169692.0000

Threshold voltage group cell usage:

>> saed32cell_svt 100.00%

Beginning Constant Register Removal

0:00:04	807.7	0.00	0.0	0.0		38169692.0000
0:00:04	807.7	0.00	0.0	0.0		38169692.0000

Threshold voltage group cell usage:

>> saed32cell_svt 100.00%

Beginning Global Optimizations

Global Optimization (Phase 1)

Global Optimization (Phase 2)

Global Optimization (Phase 3)

Global Optimization (Phase 4)

Global Optimization (Phase 5)

Global Optimization (Phase 6)

Global Optimization (Phase 7)

Global Optimization (Phase 8)

Global Optimization (Phase 9)

Global Optimization (Phase 10)

Global Optimization (Phase 11)

Global Optimization (Phase 12)

Global Optimization (Phase 13)

Global Optimization (Phase 14)

Global Optimization (Phase 15)

Global Optimization (Phase 16)

Global Optimization (Phase 17)

Global Optimization (Phase 18)

Global Optimization (Phase 19)

Global Optimization (Phase 20)

Global Optimization (Phase 21)

Global Optimization (Phase 22)

Global Optimization (Phase 23)

Global Optimization (Phase 24)

Global Optimization (Phase 25)

Global Optimization (Phase 26)

Global Optimization (Phase 27)

Global Optimization (Phase 28)

Global Optimization (Phase 29)

Global Optimization (Phase 30)

Threshold voltage group cell usage:

>> saed32cell_svt 100.00%

```
>> saed32cell_svt 100.00% Beginning Delay Optimization
```

```
>> saed32cell_svt 100.00%
```

Beginning WLM Backend Optimization

[illegible]

[illegible]

0:00:04	688.5	0.00	0.0	0.0	19313640.0000
0:00:04	688.5	0.00	0.0	0.0	19313640.0000
0:00:04	688.5	0.00	0.0	0.0	19313640.0000
0:00:04	688.5	0.00	0.0	0.0	19313640.0000
0:00:04	688.5	0.00	0.0	0.0	19313640.0000
0:00:04	688.5	0.00	0.0	0.0	19313640.0000

TOTAL						
ELAPSED	WORST NEG			SETUP	DESIGN	LEAKAGE
TIME	AREA	SLACK	COST	RULE	COST	ENDPOINT POWER

0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000
0:00:05	688.5	0.00	0.0	0.0		19313640.0000

CPU Load: 6%, Ram Free: 1 GB, Swap Free: 30 GB, Work Disk Free: 890 GB, Tmp Disk Free: 30 GB

Loading db file

'/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db'

Note: Symbol # after min delay cost means estimated hold TNS across all active scenarios

Optimization Complete

Information: State dependent leakage is now switched from off to on.

Information: Propagating switching activity (low effort zero delay simulation). (PWR-6)

CPU Load: 6%, Ram Free: 1 GB, Swap Free: 30 GB, Work Disk Free: 890 GB, Tmp Disk Free: 30 GB

Information: Total number of MV cells in the design.

MV Cells	Total Number

Level Shifter:	0
Enable Level Shifter:	0
Isolation Cell:	0
Retention Cell:	0

Retention Clamp Cell: 0
Switch Cell: 0
Always-On Cell: 0
Repeater Cell: 0

Unmapped MV Cells

0 Isolation Cells are unmapped
0 Retention Clamp Cells are unmapped

----- 1
Current design is 'uart_top'.
dc_shell> report_area

Report : area
Design : uart_top
Version: W-2024.09-SP3
Date : Mon Aug 10 11:24:11 2026
***** Library(s)
Used:

saed32rvt_tt0p85v25c (File:
/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db)

Number of ports: 22
Number of nets: 220
Number of cells: 198
Number of combinational cells: 144
Number of sequential cells: 54
Number of macros/black boxes: 0
Number of buf/inv: 31
Number of references: 22

Combinational area: 304.210372
Buf/Inv area: 43.204481
Noncombinational area: 384.265735
Macro/Black Box area: 0.000000
Net Interconnect area: 126.013993

Total cell area: 688.476107

Total area: 814.490100

1 dc_shell>

report_qor

Report : qor

Design : uart_top

Version: W-2024.09-SP3

Date : Mon Aug 10 11:24:25 2026

Timing Path Group (none)

Levels of Logic: 0.00

Critical Path Length: 0.19

Critical Path Slack: uninit

Critical Path Clk Period: 2.00

Total Negative Slack: 0.00

No. of Violating Paths: 0.00

Worst Hold Violation: 0.00

Total Hold Violation: 0.00

No. of Hold Violations: 0.00 -----

--

Cell Count

Hierarchical Cell Count: 0

Hierarchical Port Count: 0

Leaf Cell Count: 198

Buf/Inv Cell Count: 31

Buf Cell Count: 5

Inv Cell Count: 26

CT Buf/Inv Cell Count: 0

Combinational Cell Count: 144

Sequential Cell Count: 54

Macro Count: 0

Area

Combinational Area: 304.210372

Noncombinational Area: 384.265735 Buf/Inv Area: 43.204481

Total Buffer Area: 10.17

Total Inverter Area: 33.04

Macro/Black Box Area: 0.000000

Net Area: 126.013993

Cell Area: 688.476107

Design Area: 814.490100

Design Rules

Total Number of Nets: 220

Nets With Violations: 0

Max Trans Violations: 0

Max Cap Violations: 0

Hostname: taradash.soa.ac.in

Compile CPU Statistics

Resource Sharing: 0.00

Logic Optimization: 0.10

Mapping Optimization: 0.58 -----

----- Overall

Compile Time: 7.99

Overall Compile Wall Clock Time: 8.65

Design WNS: 0.00 TNS: 0.00 Number of Violating Paths: 0

Design (Hold) WNS: 0.00 TNS: 0.00 Number of Violating Paths: 0

1

```
dc_shell> get_cells
```

```
{BG/baud_tick_reg BG/count_reg[14] BG/count_reg[13] BG/count_reg[12] BG/count_reg[11]
BG/count_reg[10] BG/count_reg[9] BG/count_reg[8] BG/count_reg[7] BG/count_reg[6]
BG/count_reg[5] BG/count_reg[4] BG/count_reg[3] BG/count_reg[2] BG/count_reg[1]
BG/count_reg[0] TX/tx_reg TX/shift_reg_reg[0] TX/shift_reg_reg[1] TX/shift_reg_reg[2]
TX/shift_reg_reg[3] TX/shift_reg_reg[4] TX/shift_reg_reg[5] TX/shift_reg_reg[6]
TX/shift_reg_reg[7] TX/bit_count_reg[1] TX/bit_count_reg[0] TX/busy_reg TX/state_reg[1]
TX/bit_count_reg[2] TX/state_reg[0] RX/rx_done_reg RX/rx_data_reg[0] RX/shift_reg_reg[0]
RX/rx_data_reg[1] RX/shift_reg_reg[1] RX/rx_data_reg[2] RX/shift_reg_reg[2] RX/rx_data_reg[3]
RX/shift_reg_reg[3] RX/rx_data_reg[4] RX/shift_reg_reg[4] RX/rx_data_reg[5]
RX/shift_reg_reg[5] RX/rx_data_reg[6] RX/shift_reg_reg[6] RX/rx_data_reg[7]
RX/shift_reg_reg[7] RX/bit_count_reg[1] RX/state_reg[2] RX/state_reg[1] RX/state_reg[0]
RX/bit_count_reg[2] RX/bit_count_reg[0] U123 U124 U125 U126 U127 U128 U129 U130 U131
U132 U133 U134 U135 U136 U137 U138 U139 U140 U141 U142 U143 U144 U145 U146 U147
U148 U149 U150 U151 U152 U153 U154 U155 U156 U157 U158 U159 U160 U161 U162 U163
U164 U165 U166 U167 U168 ...} dc_shell> sizeof_collection [get_cells ]
```

198

```
dc_shell> sizeof_collection [all_registers ]
```

54

```
dc_shell> all_registers
```

```
{RX/bit_count_reg[0] RX/bit_count_reg[2] RX/state_reg[0] RX/state_reg[1] RX/state_reg[2]
RX/bit_count_reg[1] RX/shift_reg_reg[7] RX/rx_data_reg[7] RX/shift_reg_reg[6]
RX/rx_data_reg[6] RX/shift_reg_reg[5] RX/rx_data_reg[5] RX/shift_reg_reg[4] RX/rx_data_reg[4]
RX/shift_reg_reg[3] RX/rx_data_reg[3] RX/shift_reg_reg[2] RX/rx_data_reg[2]
RX/shift_reg_reg[1] RX/rx_data_reg[1] RX/shift_reg_reg[0] RX/rx_data_reg[0] RX/rx_done_reg
TX/state_reg[0] TX/bit_count_reg[2] TX/state_reg[1] TX/busy_reg TX/bit_count_reg[0]
TX/bit_count_reg[1] TX/shift_reg_reg[7] TX/shift_reg_reg[6] TX/shift_reg_reg[5]
TX/shift_reg_reg[4] TX/shift_reg_reg[3] TX/shift_reg_reg[2] TX/shift_reg_reg[1]
TX/shift_reg_reg[0] TX/tx_reg BG/count_reg[0] BG/count_reg[1] BG/count_reg[2]
BG/count_reg[3] BG/count_reg[4] BG/count_reg[5] BG/count_reg[6] BG/count_reg[7]
BG/count_reg[8] BG/count_reg[9] BG/count_reg[10] BG/count_reg[11] BG/count_reg[12]
BG/count_reg[13] BG/count_reg[14] BG/baud_tick_reg}
```

```
dc_shell> write_file -format verilog -hierarchy -output indv_synthesized.sv
```

Writing verilog file

'/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/indv_synthesized.sv'.

1

```
dc_shell> write_file ddc -hierarchy -output indv_pending.ddc
```

Warning: Can't find in memory specified design or design file 'ddc'. (UID-26)

No designs to write

0

```
dc_shell> write_file -format verilog -hierarchy -output indv_synthesized.ddc
```

Writing verilog file

'/home/iter_work/iter4/upaasana_dash/UART_sysnthesis/sysnthesis/indv_synthesized.ddc'. 1

```
dc_shell> write_sdc fa_constraints.sdc
```

1

dc_shell> report_power

Report : power

-analysis_effort low Design

: uart_top

Version: W-2024.09-SP3

Date : Mon Aug 10 11:29:25 2026

Library(s) Used:

saed32rvt_tt0p85v25c (File:
/home/iter_work/KIT/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_tt0p85v25c.db)

Operating Conditions: tt0p85v25c Library: saed32rvt_tt0p85v25c

Wire Load Model Mode: enclosed

Design	Wire Load Model	Library -----
----- uart_top		
8000	saed32rvt_tt0p85v25c	

Global Operating Voltage = 0.85 Power-specific

unit information :

Voltage Units = 1V

Capacitance Units = 1.000000ff

Time Units = 1ns

Dynamic Power Units = 1uW (derived from V,C,T units)

Leakage Power Units = 1pW

Attributes -----

i - Including register clock pin internal power

Cell Internal Power = 7.7455 uW (81%)

Net Switching Power = 1.7611 uW (19%)

Total Dynamic Power = 9.5066 uW (100%)

Cell Leakage Power = 16.7743 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

Power Group	Internal Power	Switching Power	Leakage Power	Total Power (%)	Attrs -----

io_pad	0.0000	0.0000	0.0000	0.0000 (0.00%)	memory
0.0000	0.0000	0.0000	0.0000 (0.00%)	black_box	0.0000
0.0000	0.0000	0.0000 (0.00%)	clock_network	0.0000	0.0000
0.0000	0.0000 (0.00%)	i register	0.0000	0.0000	0.0000
0.0000 (0.00%)	sequential	6.5973	7.4641e-02	1.2380e+07	19.0524
(72.50%)	combinational	1.1481	1.6865	4.3938e+06	7.2284 (27.50%)

Total	7.7455 uW	1.7611 uW	1.6774e+07 pW	26.2809 uW	1

7. Static Timing Analysis reports

I. For command compile:

dc_shell> report_timing

Information: Updating design information... (UID-85)

Report : timing

-path full

-delay max

-max_paths 1

Design : uart_top

Version: W-2024.09-SP3

Date : Mon Aug 10 11:22:09 2026

Operating Conditions: tt0p85v25c Library: saed32rvt_tt0p85v25c

Wire Load Model Mode: enclosed

Startpoint: TX/tx_reg (rising edge-triggered flip-flop)

Endpoint: tx (output port clocked by clk)

Path Group: (none)
Path Type: max

Des/Clust/Port	Wire Load Model	Library	-----
		uart_tx	
8000	saed32rvt_tt0p85v25c	uart_top	
8000	saed32rvt_tt0p85v25c		

Point	Incr	Path	

TX/tx_reg/CLK (DFFASX1_RVT)	0.00	0.00	r
TX/tx_reg/Q (DFFASX1_RVT)	0.19	0.19	r
TX/tx (uart_tx)	0.00	0.19	r tx (out)
0.00	0.19	r	data arrival time
0.19			

1

II. For command compile ultra:

dc_shell> report_timing
Information: Updating design information... (UID-85)

Report : timing
-path full
-delay max
-max_paths 1 Design

: uart_top
Version: W-2024.09-SP3
Date : Mon Aug 10 11:22:40 2026

Operating Conditions: tt0p85v25c Library: saed32rvt_tt0p85v25c
Wire Load Model Mode: enclosed
Startpoint: TX/tx_reg (rising edge-triggered flip-flop)
Endpoint: tx (output port clocked by clk)
Path Group: (none)
Path Type: max

Des/Clust/Port	Wire Load Model	Library	-----
8000	saed32rvt_tt0p85v25c	uart_top	
Point	Incr	Path	
TX/tx_reg/CLK (DFFASX1_RVT)		0.00	0.00 r
TX/tx_reg/Q (DFFASX1_RVT)		0.19	0.19 r
tx (out)	0.00	0.19 r	data
arrival time		0.19	
			----- 1

8. Conclusions

The 8-bit UART system was successfully designed and implemented using Verilog HDL. The design consists of a baud-rate generator, UART transmitter, UART receiver, and top-level UART module. The transmitter and receiver use FSM-based control to ensure proper UART frame generation and reception.

The design was functionally verified using dedicated testbenches and multiple data patterns, with waveform analysis confirming the expected operation of the UART signals and data transfer. The complete RTL design was then prepared for synthesis using Synopsys Design Compiler, enabling analysis of its area, timing, power, and cell utilization.

Overall, the project demonstrates the complete digital design flow from **RTL development and functional verification to synthesis and timing analysis**, providing a practical implementation of an 8-bit UART communication system.